

PAPER_676: UQFF Analysis of GW190425: Heaviest Known NS Binary and Ejecta Constraints

Author: Daniel Murphy

Framework: Unified Quantum Field Framework (UQFF) v5.30

Session: 173

Date: April 1, 2026

Class: UQFFComparedToGW190425 | CP4 #260

Source: grok_share_fc21e30c24b4.txt

Domain: Gravitational Waves / Dense Matter

Abstract

GW190425 represents the heaviest binary neutron star system observed (total mass $3.4 M_{\odot}$). The UQFF framework predicts a post-merger phase shift $\phi_{\text{UQFF}} = \kappa f_{\text{TRZ}} t_{\text{merger}}$ and constrains the ejecta mass through the condensate suppression: $M_{\text{ej,UQFF}} < M_{\text{ej,GR}} \cdot (\rho_{\text{SCm}}/\rho_{\text{UA}})(1 - f_{\text{TRZ}})$, resulting in an ejecta upper limit nearly two orders of magnitude below GR.

Primary UQFF Equation

$$M_{\text{ej,UQFF}} = 0.05 M_{\text{tot}} \cdot \frac{\rho_{\text{SCm}}}{\rho_{\text{UA}}} (1 - f_{\text{TRZ}})$$

Parameters

Parameter	Value	Description
m_1	$1.9 M_{\odot}$	NS1 mass
m_2	$1.5 M_{\odot}$	NS2 mass
d	159 Mpc	Distance

UQFF Constants

Constant	Value	Description
ρ_{UA}	$7.09 \times 10^{-36} \text{ kg/m}^3$	Universal Aether density
ρ_{SCm}	$7.09 \times 10^{-37} \text{ kg/m}^3$	Schwarzschild condensate
f_{TRZ}	0.1	Time-reversal zone factor
κ	$5 \times 10^{-4} \text{ day}^{-1}$	UQFF calibration constant
[SSq]	0.57	Superstring quenching factor
μ_J	$3.38 \times 10^{23} \text{ J}\cdot\text{m}$	Magnetic string coupling
γ	$5 \times 10^{-5} / 86400 \text{ s}^{-1}$	Aether oscillation decay

C++ Implementation

```
#include "UQFFComparedToGW190425.h"
```

```
UQFFComparedToGW190425 obj;  
auto result = obj.compute_primary(...); // primary equation  
obj.simulate_over_M(M_start, M_end, dM, "output.csv");
```

Python CP4 Calculator

```
from CondensedPhysics2 import UQFFComparedToGW190425Calculator
```

```
calc = UQFFComparedToGW190425Calculator()  
result = calc.compute() # returns all UQFF quantities  
print(result)
```

References

Abbott et al. 2020 (GW190425); UQFF Session 173

UQFF v5.30 | Session 173 | April 1, 2026 | PAPER_676 of 1000